Sugar Content Estimation and Cola Preference Study

AUTHOR
Dylan Renard

PUBLISHED
August 15, 2025

1 Abstract

We examined gender differences in **Coke vs Pepsi** preference and the accuracy of sugar estimates for **12-oz** sodas in a randomized-order, convenience-sample survey. Outliers (>100 g on any can) were excluded *a priori*. We used **prop.test**, **paired t-tests** with **Wilcoxon** sensitivity, and **Welch/Mann–Whitney** for group comparisons of standardized z-scores and absolute error. We add per-hypothesis p-values, confidence intervals, explicit decision statements, societal implications, and brief power notes.

2 Introduction

Public perception of beverage sugar content often diverges from nutritional values, potentially influencing consumption behavior. We assessed (a) whether Coke is equally preferred across gender and (b) whether branding shifts perceived sugar content relative to an unbranded control. Secondary analyses explored standardized within-person sweetness sentiment and estimation accuracy.

3 Method

3.1 Participants and Sampling

Participants were recruited via a **convenience sample** from my workplace, University of Iowa researchers, Johns Hopkins Bioinformatics students, and LinkedIn connections. The sample exhibited slightly higher male participation and a skew toward Coke preference.

3.2 Materials and Procedure

Participants viewed five can images—Silver (Q1), Blue (Q2), Red (Q3), Pepsi (Q4), and Coke (Q5)—in **randomized order** and estimated grams of sugar for each. Nutritional labels were hidden. Cola preference, weekly intake, age, and gender identity were collected.

Figure 1. Images presented to participants (randomized order).



Silver (Q1)



Blue (Q2)



Red (Q3)



Pepsi (Q4)



Coke (Q5)

3.3 Design and Bias Control

This was an **observational** survey with repeated measures per participant (Q1–Q5). To mitigate order effects, **image order was randomized** uniformly. Branding remained visible for Coke/Pepsi; the Silver can acted as a neutral control.

3.3.1 Blinding Justification

A **fully blinded taste test** or label-masking paradigm would better isolate branding effects. Given logistical constraints (remote data collection, single-session design), complete blinding was not feasible. We therefore used **randomized image order** and a **neutral (Silver) control** as practical bias controls. This choice is acknowledged as a limitation.

3.4 Measures

- **Sugar estimates:** self-reported grams for each can.
- **Preference:** Coke/Pepsi/Diet.

- **Weekly intake:** numeric, used to classify **Soda vs Non-soda** drinkers.
- **Demographics:** age, gender identity (consolidated into Female/Male/Other for analysis).

3.5 Data Treatment

Outliers were defined *a priori* as any record with **any** sugar guess **> 100 g**, removed to protect estimates from extreme entries. This threshold is conservative relative to typical 12-oz soda values and was established before hypothesis testing. No personal identifiers are reported.

3.6 Sample Size and A Priori Power

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N_total_raw	N_removed_outliers_gt100g	N_analytic
62	5	57

Table 1. Sample size counts.

For **H1** (proportion test within each gender), power to detect a **15-point deviation** (0.65 vs 0.50) at α = .05 was approximately **Female:** 18%, **Male:** 23% (Pagano & Gauvreau, 2018). For **H2** (paired), given the observed SD of differences 20.43 and 57 pairs, the smallest **detectable difference** with ~80% power is about **7.8 g**.

4 Results

4.1 Descriptive Statistics

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gender_simple	n	pct
Female	24	42.1
Male	32	56.1
Other	1	1.8

Table 2. Descriptive summaries.

▶ Code

prefer	n
Coke	45

prefer	n
Pepsi	12

Table 2. Descriptive summaries.

▶ Code

	Var1	Var2	Freq
1	A	Min.	18
2	A	1st Qu.	24
3	A	Median	27
4	A	Mean	29.36842105263158
5	A	3rd Qu.	31
6	A	Max.	58

Table 2. Descriptive summaries.

4.2 H1 — Coke vs Pepsi Preference by Gender (Primary)

Hypothesis. Within each gender, P(Coke) = .50 (two-sided).

Method. One-sample **prop.test** against .50 within Female and Male groups.

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gender	n	coke_n	prop_coke	p_value	ci_l	
Female	24	18	0.75	0.01430587843542965	0.55100555994846	0.8800063
Male	32	27	0.844	0.0001006219221196369	0.6824585040246961	0.9313557

Narrative Results. In our convenience sample drawn from my personal and professional circles, **Coke preference exceeded Pepsi in both genders**. For **Females**, the proportion preferring Coke was **0.75** (95% CI **[0.55, 0.88]**), p = **0.014** vs .50; for **Males**, the proportion was **0.844** (95% CI **[0.68, 0.93]**), p = **<0.001**. We therefore **reject H0 for Females** and **reject H0 for Males**. Societally, if this preference pattern holds beyond our sample, it suggests brand loyalty toward Coke across genders within similar communities, relevant to **marketing strategies** and potential **public-health messaging** around sugary beverage choice. Notably, several participants in my circle volunteered that they “usually choose Coke,” aligning with the observed proportions.

4.3 H2 — Silver vs Coke Estimates (Paired, Primary)

Hypothesis. Mean difference (Q5 − Q1) = 0 (two-sided).

Method. Paired t-test on within-participant differences (Coke − Silver); **Wilcoxon** as sensitivity. Normality was visually assessed via histograms and QQ plots.

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metric	value
Mean (Q5 − Q1)	16.80701754385965
Paired t p	6.845557109140549e-8
95% CI (L)	11.38744715768876
95% CI (U)	22.22658793003054
Wilcoxon p	2.947206161744969e-8

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Figure 2. Histogram of within-person differences (Coke − Silver). Bars to the right indicate higher perceived sugar for Coke. Useful to check approximate symmetry for t-test assumptions.
Figure 3. QQ plot of (Coke − Silver) differences. Points close to the diagonal indicate approximate normality; large departures motivate nonparametric sensitivity (Wilcoxon).
Figure 4. Boxplot of raw estimates for Silver vs Coke across participants. The median line and IQR convey central tendency and variability of perceived sugar grams.
Narrative Results. The mean within-person difference (Coke − Silver) was **16.81 g** (95% CI **[11.39, 22.23]**). The paired **t-test** yielded p = **<0.001**, with Wilcoxon **p = <0.001** confirming the pattern. We therefore **reject H0 (branding shifts estimates)** at α = .05. If this difference generalizes, it implies **branding may systematically bias perceived sugar**, a consideration for **consumer education** and **labeling policy**. However, given convenience sampling and limited N, these inferences should be validated in larger, more controlled studies.

4.4 H3 — Relative Sweetness Sentiment (*z*) Across Groups (Secondary)

Hypothesis. Mean *z*-scores equal across groups (two-sided).

Method. Standardized within-person *z*-scores across Q1–Q5; compared groups via **Welch t** and **Mann–Whitney**.

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variable	contrast	n	welch_p

q1_silver_z	Non-soda vs Soda	49	0.6642399910772119	0.339558811448784
q5_coke_z	Non-soda vs Soda	49	0.4025019805973687	0.4413147366721604
q1_silver_z	Female vs Male	48	0.3798089947183587	0.9669580893827953
q5_coke_z	Female vs Male	48	0.9556673588666518	0.6712257298716511
q1_silver_z	Diet vs Coke	38		
q5_coke_z	Diet vs Coke	38		

Narrative Results. Group comparisons of standardized sweetness sentiment yielded mixed evidence. For example, **Non-soda vs Soda** on Coke (*z*) produced Welch p = **0.403** and Mann–Whitney p = **0.441**, leading us to **fail to reject H0**. If significant, such differences suggest **habitual soda intake may shape relative sweetness expectations**, relevant for designing **targeted nutrition education**. Comparisons by gender (**Female vs Male**) and preference type (**Diet vs Coke**) show similar decision logic; **non-significant** results likely reflect **limited subgroup sizes** rather than true equivalence.

4.5 H4 — Coke Absolute Error Across Groups (Secondary)

Hypothesis. Mean |Coke guess − 39| equal across groups (two-sided).

Method. Group comparisons using **Welch t** and **Mann–Whitney**.

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variable	contrast	n	welch_p	
coke_abserr	Non-soda vs Soda	57	0.5526015865813904	0.2464078037814602
coke_abserr	Female vs Male	56	0.01176244045767471	0.01868148880161355
coke_abserr	Diet vs Coke	45		

Narrative Results. Absolute error in Coke estimates reflects **practical accuracy**. For **Non-soda vs Soda**, Welch p = **0.553** and Mann–Whitney p = **0.246** lead us to **fail to reject H0**. If significant, **non-soda drinkers** may benefit from **targeted education** on typical sugar levels because their internal references are less calibrated. Patterns by **gender** and **preference** mirror this logic; **null findings** should be interpreted cautiously due to limited power.

5 Discussion

Across hypotheses, we used standard methods from class (Pagano & Gauvreau, 2018) to evaluate preference balance, branding effects, relative sentiment, and accuracy. In our circles (workplace, academic peers), **both women and men tended to prefer Coke**, aligning with the H1 estimates and suggesting stable **brand loyalty** within similar communities. When branding effects on perceived sugar are present (H2), they imply that **labels and**

brand imagery might influence estimates independent of actual content, which matters for **public-health communication**. H3–H4 patterns indicate that **soda exposure and preference** can relate to **how people interpret sweetness** and **how accurately they estimate content**; these are actionable levers for **nutrition education**.

6 Conclusions

- **H1 (preference):** We **reject H0 for Females** and **reject H0 for Males** for P(Coke) = .50; in **both genders** within my circle, **Coke was preferred** to Pepsi.
- **H2 (paired):** We **reject H0 (Coke vs Silver differ)**, consistent with potential branding influence on perceived sugar.
- **H3/H4 (secondary):** Group differences in standardized sentiment and accuracy are **mixed and power-limited**; non-significant results should be interpreted cautiously.

Public-health implication. A larger, more structured study with **pre-/post-knowledge testing** (e.g., educating participants on the sugar in an **8-oz** can between sessions) could quantify how **information changes estimation** and whether effects **persist**. Such designs would be valuable for **health agencies** developing interventions to reduce sugar overconsumption.

7 References

Pagano, M., & Gauvreau, K. (2018). *Principles of biostatistics*. Duxbury Press.